



NAME:

TEACHER:

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Level 1 Mathematics and Statistics

Practice Examination 2025

91947

Demonstrate mathematical reasoning

Credits: Five

Achievement	Achievement with Merit	Achievement with Excellence
Demonstrate mathematical reasoning.	Demonstrate mathematical reasoning with relational thinking.	Demonstrate mathematical reasoning with extended abstract thinking.

You should attempt ALL the questions in this booklet.

If you need more room for any answer, use the extra space provided at the back of this booklet.

Check that this booklet has pages 2–10 in the correct order and that none of these pages are blank.

Do not write in the grey shaded columns on the right of each page. This area is for markers only.

YOU MUST HAND THIS BOOKLET TO THE SUPERVISOR AT THE END OF THE EXAMINATION.

QUESTION ONE



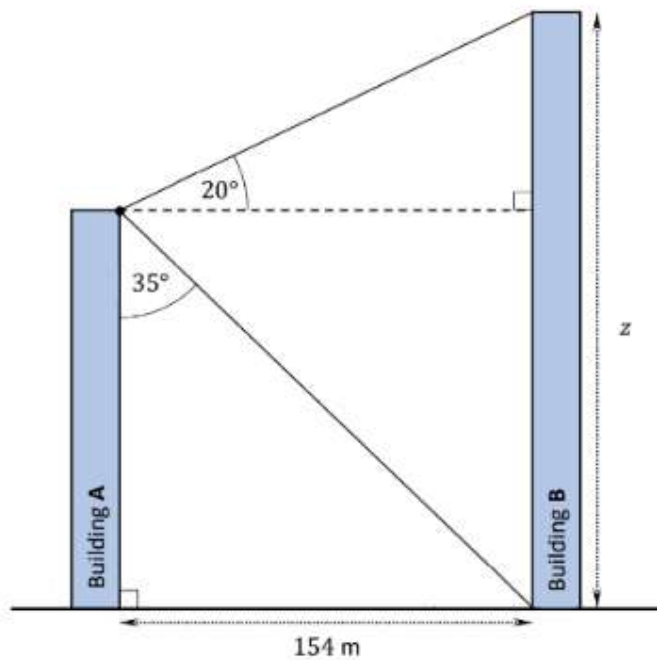
- (a) Kōtiro buys a cellphone for €399 from Europe.
The exchange rate for the NZ dollar to Euro is $\$1 = \text{€}0.53$.
Kōtiro must also pay 15% GST on this purchase price.
What is the total cost for the purchase of this cellphone in NZ dollars?

- (b) Three triangles **A**, **B** and **C** are shown below.
All lengths are measured in centimetres.

Triangle A	Triangle B	Triangle C

- i) The perimeter of triangle **B** is 24 cm.
Calculate the value of x .
- ii) The perimeters of the three triangles **A**, **B** and **C** form a **linear** pattern.
The perimeter of triangle **A** is 8 cm. Triangle **C** has the largest perimeter.
Use this information to work out the value of y in triangle **C**.

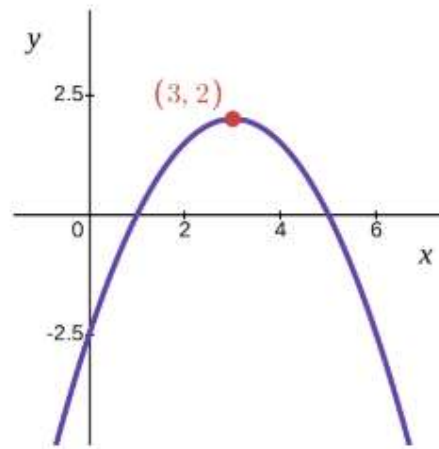
- (c) The diagram below shows two vertical buildings, **A** and **B**.
The top of building **B** is 20° from horizontal to the top of building **A**.
The bottom of building **B** is 35° from the vertical side of building **A**.
The buildings are 154 metres apart.



Calculate the value of z , the height of building **B**.

(d) State the equation of this parabola.

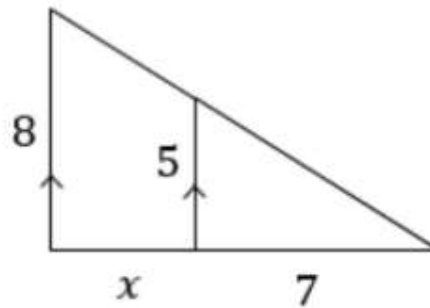
It passes through the x -axis at $(1, 0)$ and $(5, 0)$.



Assessor's
use only

QUESTION TWO

- (a) Calculate the value of x .



- (b) A sequence is shown in the table below:

x	y
1	7
2	10
3	19
4	34
5	55

- (i) Find an equation that represents y , for any given x -value. Justify your answer by showing appropriate mathematical working.

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- ii) If the above equation was graphed, it can be said that y would never equal zero. Using a sketch and features of the graph, discuss why this is true.

- (c) A rectangle has an area of 20 cm^2 , a length of $3x + 7 \text{ cm}$ and width of $x \text{ cm}$.

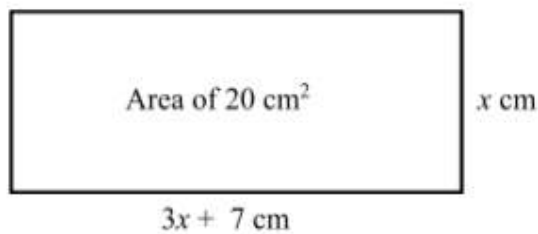


Diagram is
NOT to scale

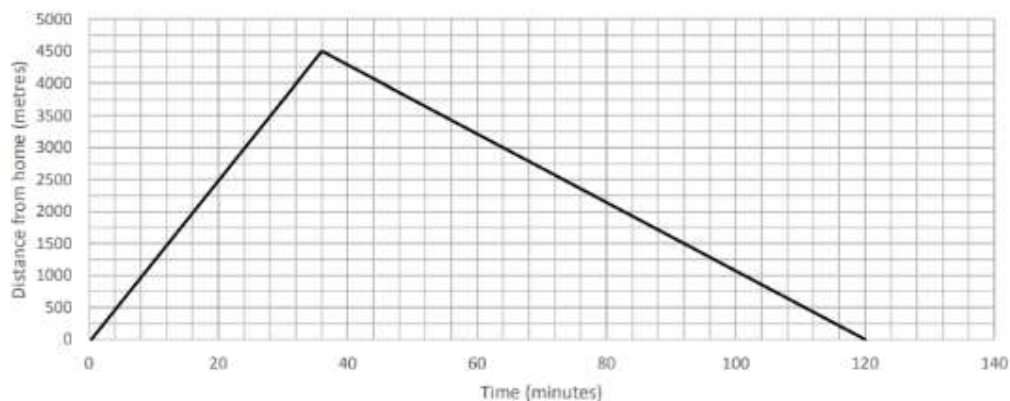
What is the value of x ?

- (d) Find the coordinates of point A .



QUESTION THREE

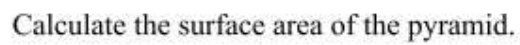
- (a) Mila went for a bike ride.
The graph below shows her journey.



Find Mila's average speed for the bike ride, in kilometres per hour.

- (b) Simplify $\frac{2x^2+5x-12}{x^2-16}$

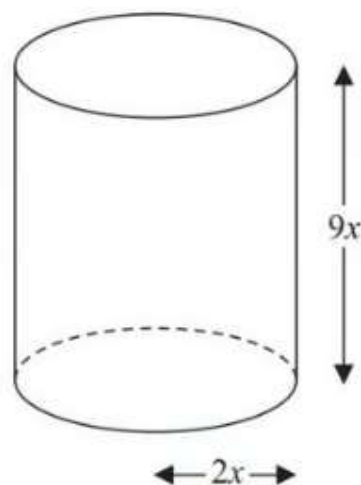
- Diagram is
NOT to scale

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

(d) Solve: $\frac{x}{2} + \frac{x-2}{3} < 5$

- (e) The diagram shows a solid metal cylinder.
The cylinder has base radius $2x$ and height $9x$.
The cylinder is melted down and made into a sphere
of radius r .

*Diagram is
NOT to scale*



Find an expression for r in terms of x for the
volume of the sphere.
